

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0680

Scott Co 100 kV
March 30, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0680. The Interconnection Customer proposes a 211.6 MW solar photovoltaic facility interconnecting at the Scott Co 100 kV substation in the Ohio area. The request was received on September 25, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 279 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,201,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Washington County Cogeneration Facility-Villa Park 230kV
Contingency: Wolfe-Barrett 500kV Line
Loading: 107.3% (Rating: 856 MVA)
DFAX: 14.8%
Pre-project loading: 103.7%
2. Facility: Villa Park-Wolfe Substation 230kV line
Contingency: Wolfe-Barrett 500kV
Loading: 103.4% (Rating: 920 MVA)
DFAX: 14.8%
Pre-project loading: 100.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Washington County Cogeneration Facility-Villa Park 230kV line	Reconductor line	\$12,128,000	7% (\$807,000)
Villa Park-Wolfe 230kV L	Reconductor line	\$5,922,000	7% (\$394,000)
	Total allocated to project		\$1,201,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.